Arthur Lincoln

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COMPUTER ENGINEERING STUDENT

CAREER OBJECTIVE

Computer Engineering student at UNIVASF seeking a software development internship. I program in C, Java and Python, with projects in data structures, search algorithms, object-oriented design and game development.

EDUCATION

B.S. Computer Engineering Federal University of the São Francisco Valley (UNIVASF) In progress
Juazeiro, Brazil

LL.B. Law Unibras In progress
Juazeiro, Brazil

SKILLS

C (C99)

Java

Python

Data structures

Object-oriented programming

Breadth-first search

raylib

Git and GitHub

GCC, Make and Maven

Linux

LANGUAGES

NATIVE
Portuguese

COURSES

UDEMY, 2026
Python 3 from basic to advanced, 141.5 h

PROJECTS

2026

Eixo 0 | Object-Oriented Programming | UNIVASF

- 2D metroidvania in Java 17 with FXGL, built with Raioni Felix and Hiago Lima: connected rooms, three enemy types, melee attack and dash.
- I built the player and its physics, gravity, enemy collision and knockback, projectiles, the menu, and levels loaded from **JSON files** with Gson.
- In the barebones version, the game was rewritten as 28 plain Java classes using abstract classes, interfaces and polymorphism.

2026

prolog-simulator-c | Algorithms and Data Structures | UNIVASF

- Built a Prolog-style interpreter with a parser for facts and rules, predicates indexed in a binary search tree and clauses stored in linked lists.
- Resolution engine with unification, variable renaming and backtracking that finds **every solution** to a query.
- Knowledge bases loaded from .txt files. Pair project with Caio Benevides.

2026

maze-solver-bfs | Algorithms and Data Structures | UNIVASF

- Implemented breadth-first search with a sequential queue plus visited and parent matrices to rebuild **the shortest path** to the exit.
- Step-by-step terminal animation with ANSI colors; loads 10 mazes from text files. Pair project with Caio Benevides.

2025

raylib-farm | Algorithms and Programming | UNIVASF

- Built a 2D farming game in C with raylib: scrollable shop, inventory, time-based crop growth on a grid, and an XP and level system.
- Shipped a browser version (WebAssembly) and Windows and Linux builds that **don't require installing raylib**.
- While revising the project, fixed 4 bugs, including one that **allowed infinite money**.

OTHER PROJECTS

ELECTROTECHNICS

EMBEDDED IRRIGATION SIMULATOR IN C | UNIVASF, with Hiago de Oliveira Lima and Rafaela Nunes Albuquerque